

Compendium of Theorems

of the Beating Substrate Theory

Complete and optimized version — with formalizability classification

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*“Another entry door to BST is its compendium of theorems.
They do not reinvent Physics but allow rejoining it
from the deepest structures of the theory.”*

*Computational assistance was used for the organization and consistency auditing of this compendium.
Every stage attribution was verified against the repository.*

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Chapter 1

Reading Note

1.1 What this compendium is, and is not

This compendium gathers, as numbered statements, the structural results of the Beating Substrate Theory (BST), in their most complete and optimized form to date. In keeping with the epigraph, these theorems do not reinvent physics: they allow one to *rejoin* it from the deepest structures of the framework.

This compendium is not a proof that nature *is* the beating substrate. The physical theorems *rejoin* equations that are already established and already calibrated against experiment; they do not derive them *ex nihilo* and do not constitute new empirical confirmation. The numerical results are *possibility/consistency* results (the substrate–phase language *can* host these structures), not measurements of nature. No new, falsified prediction is claimed.

1.2 The two doors — and what their convergence is worth

BST was reached by two independent paths: a *constructive* approach (building a simulation honouring observed reality, then reading off its structure) and a *re-description* approach (starting from established physics — already calibrated — and reformulating it under a minimal geometric principle). The two converge on the same architecture. This convergence attests to the *robustness* and *non-arbitrariness* of the re-description — but not to independent empirical validation: both doors honour the same anchor (known physics), and a simulation reproduces only what it was built to reproduce. It is an argument of coherence, not a measurement of nature.

1.3 Status legend

Status	Meaning
Forced	Imposed by the structure of the framework, with no free tuning.
Derived	Obtained by derivation/demonstrative chain under explicit hypotheses (possibility result).
Conditional	Valid if an explicitly posited ingredient is accepted.
Posited	Explicitly assumed input, not derived (axiomatic boundary).
Calibrated	Magnitude tied to observation; legitimate when declared.
Consistent	Compatible with the framework, but not derived.
Open	Question unresolved by the current framework.
Exploratory	Proposed conceptual architecture, not anchored as the physics is.
Methodological	Constraint imposed on the framework, not derived from it.

1.4 Formalizability legend

Index	Meaning
$\vdash_L$	Logical-structural: formalizable as an implication, once its primitives are defined.
$\vdash_P$	Standard physics: its proof belongs to known mathematics and does not validate BST.
$\vdash_E$	Exploratory: primitives not mathematically defined; not formalizable as it stands.

Lean caveat. A proof assistant checks logical validity *given axioms*, not correspondence with the world. A verified $\vdash_L$ theorem strengthens the internal coherence of the framework; it establishes neither empirical truth, nor the physical calibrations, nor the cognitive layer.

Chapter 2

Ontological Theorems

Theorem 2.1 (Existence). There is something rather than nothing; the minimal irreducible starting point.

Status: Forced (foundational axiom) | *Formalizability:* $\vdash_L$ | *Depends on:* — | *Stage(s):* —

Theorem 2.2 (Substrate Necessity). A persistent organization requires a support: the latent beating substrate.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Existence | *Stage(s):* —

Theorem 2.3 (Beating Necessity). The minimal dynamics take the form of resolution-dependent beatings; no single universal beating is assumed.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Substrate | *Stage(s):* 007, 009

Theorem 2.4 (Closure $\Rightarrow$ Stability). A stable organization requires closure; stability emerges from closure + persistence.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Beatings, Closure | *Stage(s):* 067

Chapter 3

Theorems of Resolution and Latency

Theorem 3.1 (Resolution Dependence). Every reconstruction is relative to a resolution.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Stability | *Stage(s):* 008

Theorem 3.2 (Latent Depth). The substrate carries depth beyond the observable cutoff; the latent can control the observable.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Resolution | *Stage(s):* 004, 010–013

Theorem 3.3 (Inter-Resolution Persistence). A sufficiently coherent structure remains reconstructible across multiple resolutions.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Resolution | *Stage(s):* 007

Theorem 3.4 (Latent Necessity). Under limited resolutions + observable reconstruction + inter-resolution persistence, the existence of a latent reconstruction is a *necessary* consequence — not a postulate.

Status: Derived | *Formalizability:* $\vdash_L$ | *Depends on:* Observable, Persistence | *Stage(s):* 004

Note. The cleanest candidate for formalization: an implication between precise conditions $\Rightarrow$ non-emptiness of the latent complement.

Chapter 4

Reconstruction Theorems

Theorem 4.1 (Identity). An identity emerges from a persistent reconstruction, coherent and continuous across time and resolution; a reconstructive property, not a substance.

Status: Derived | *Formalizability:* $\vdash_L$ (defs. to be made precise) | *Depends on:* Persistence | *Stage(s):* —

Theorem 4.2 (Memory). Preserving reconstructive traces that influence later reconstruction requires a persistent identity.

Status: Derived | *Formalizability:* $\vdash_L$ (defs. to be made precise) | *Depends on:* Identity | *Stage(s):* —

Theorem 4.3 (Reconstruction). Observable and latent reconstruction are two aspects of a single operation; the distinction concerns reconstructibility, not existence.

Status: Demonstrated | *Formalizability:* $\vdash_L$ | *Depends on:* Observable, Latent | *Stage(s):* 065

Note on the Formalization of Persistence. Two distinct formalizations of persistence have been investigated within the Lean verification program associated with BST.

The first identifies persistence with cross-resolution distinguishability, expressed as a global separation condition on observable states.

The second identifies persistence with cross-resolution coherence, expressed as a gluing property between compatible local representations.

Formally verified counterexamples show that these two notions are logically independent. Consequently, coherence cannot be reduced to distinguishability.

This observation is consistent with the interpretation adopted throughout the main manuscript, according to which what persists across resolutions is the coherence of the reconstructed organization rather than the preservation of any particular local form.

Chapter 5

Cognitive Theorems (Exploratory)

*Epistemic status of this entire chapter: Exploratory.***Theorem 5.1 (Imagination).** A memory-bearing reconstruction can generate coherent organizations not constrained by present observation.

Status: Exploratory | *Formalizability:* $\vdash_E$ | *Depends on:* Memory | *Stage(s):* —

Theorem 5.2 (Selection). When the space of possibilities exceeds what can remain coherent, a reduction emerges through closure, stability, memory.

Status: Exploratory | *Formalizability:* $\vdash_E$ | *Depends on:* Imagination | *Stage(s):* —

Theorem 5.3 (Arbitration). When several admissible reconstructions remain, a resolution step (arbitration) produces the effective reconstruction.

Status: Exploratory | *Formalizability:* $\vdash_E$ | *Depends on:* Selection | *Stage(s):* —

Theorem 5.4 (Agent). A persistent organization operating through memory, imagination, selection, arbitration can behave as an agent. No substance/soul/hidden entity is posited; the libertarian reading of choice is *posited*, not proven.

Status: Exploratory | *Formalizability:* $\vdash_E$ | *Depends on:* Identity, Cognitive arch. | *Stage(s):* —

Chapter 6

Physical Theorems — Gauge and Electromagnetism

These theorems *rejoin* established physics; $\vdash_P$ recalls that their proof belongs to known mathematics. **Theorem 6.1 (Compact U(1) Gauge Sector)**. Plaquette, flux tube and effective

monopole provide a gauge-like reconstruction from the phase gradient.

Status: Derived (reconstruction example) | *Formalizability:* $\vdash_P$ | *Depends on:* Phase gradient | *Stage(s):* 001–003

Theorem 6.2 (Polarity and Coulomb Residual). Phase polarity and cancellation/residual reconstruct a Coulomb-like sector.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Phase gradient | *Stage(s):* 025, 026

Theorem 6.3 (Long-Range Goldstone). A broken symmetry generates a long-range Goldstone mode.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Polarity | *Stage(s):* 027

Theorem 6.4 (Magnetism as Vorticity). Magnetism (the Lorentz force) reconstructs as vorticity of the phase field.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Phase gradient | *Stage(s):* 059

Theorem 6.5 (Light as Goldstone Wave). Light reconstructs as a transverse Goldstone wave; classical electromagnetism (Maxwell) is rejoined.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Goldstone | *Stage(s):* 060

Equation rejoined: Native field $A_{\text{eff}} \sim \nabla\theta$; classical Maxwell equations.

Chapter 7

Physical Theorems — Criticality, Matter, Quantum

Theorem 7.1 (KT Criticality). The phase system undergoes a Kosterlitz–Thouless transition: unbinding of topological defects and universal critical invariants.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Phase dynamics | *Stage(s):* 031, 032, 033

Equation rejoined: $C(r) \sim r^{-\eta}$, $\eta = \frac{1}{4}$; universal jump $\rho_s = \frac{2}{\pi}$; fixed point $K_R^* = \frac{2}{\pi}$.

Theorem 7.2 (Matter as Soliton). Matter emerges as a stable topological organization (kink/lump, vortex), not as a primitive substance.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Criticality | *Stage(s):* 036

Equation rejoined: Mass as a sine–Gordon kink; $M = 8$ (model units).

Theorem 7.3 (Quantum Potential). The geometry of the phase field generates a term equivalent to the quantum potential.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Matter | *Stage(s):* 029

Equation rejoined: $Q \propto -\frac{\nabla^2 R}{R}$.

Theorem 7.4 (de Broglie Relations). Stable propagating structures satisfy the de Broglie relations.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Quantum | *Stage(s):* 037

Equation rejoined: $p = \hbar k$, $E = \hbar\omega$, $\lambda = h/p$, $v_{\text{ph}}v_{\text{gr}} = c^2$.

Theorem 7.5 (Schrödinger Limit). In the dilute limit, the effective dynamics are linear and of Schrödinger type.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Quantum | *Stage(s):* 048

Equation rejoined: $i\hbar \partial_t \psi = \hat{H}\psi$.

Theorem 7.6 (Born Consistency). $P \propto |\psi|^2$ is compatible with the framework; adopted, not derived.

Status: Consistent / Calibrated | *Formalizability:* $\vdash_P$ | *Depends on:* Quantum | *Stage(s):* 038

Chapter 8

Physical Theorems — Fermions and Spin

Theorem 8.1 (Fermions by Flux Attachment). Spin and fermionic statistics reconstruct via flux attachment to defects.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Topological defects | *Stage(s):* 049

Theorem 8.2 (Spinor and Double Cover). A spin- $\frac{1}{2}$ spinor emerges from a minimal factorization; 4π double cover (Clifford).

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Fermions | *Stage(s):* 056

Theorem 8.3 (Dirac Operator). The walk/zitterbewegung reconstructs the Dirac equation, first in 1+1 then in 3+1, with $H^2 = p^2 + m^2$.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Spinor, Dispersion | *Stage(s):* 051, 071

Equation rejoined: $H^2 = p^2 c^2 + m^2 c^4$; mass as inversion rate (zitterbewegung).

Chapter 9

Physical Theorems — Relativity and Gravity

Theorem 9.1 (Relativistic Dispersion). Conservative beating dynamics generate a relativistic dispersion and a limiting speed c .

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Quantum | *Stage(s):* 034

Equation rejoined: $\omega^2 = c^2 k^2 + m^2 c^4 / \hbar^2$; $E^2 = p^2 c^2 + m^2 c^4$.

Theorem 9.2 (Effective Metric). An inhomogeneous substrate organization modifies propagation geometrically: an effective acoustic metric.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Relativistic | *Stage(s):* 035

Theorem 9.3 (Induced Gravity and Newton). Curvature from persistent structures induces gravity (Sakharov), reproducing Newton at long range.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Effective metric | *Stage(s):* 039–041

Equation rejoined: $F \propto \frac{M_1 M_2}{r^2}$; $G_{\text{eff}} = \frac{\lambda^2}{4\pi}$.

Theorem 9.4 (Light Bending). Massless excitations undergo gravitational deflection (lensing).

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Effective metric | *Stage(s):* 042

Theorem 9.5 (Conical Einstein in 2+1). The disclination cone reconstructs exactly Einstein gravity in 2+1 dimensions.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Effective metric | *Stage(s):* 045

Theorem 9.6 (Spin-2 and Einstein-Consistent Geometry). Transverse elastic shear modes generate a spin-2 sector; identified with Kleinert curvature, the framework rejoins $\gamma_{\text{PPN}} = 1$.

Status: Derived / Conditional | *Formalizability:* $\vdash_P$ | *Depends on:* Newton, Kleinert | *Stage(s):* 043, 046, 054, 057, 058

Equation rejoined: Kleinert curvature $\gamma = 1$; gravitational waves at c .

Chapter 10

Standard Model Input Theorems (Conditional)

Theorem 10.1 (Z_3 Colour). An internal Z_3 (colour) symmetry is *posited* as input; the Coleman–Mandula theorem forbids deriving it from spatial directions alone.

Status: Posited (axiomatic boundary) | *Formalizability:* $\vdash_P$ | *Depends on:* — | *Stage(s):* 069, 074–076

Theorem 10.2 (Three Generations). Three observable generations are interpreted via a resolution threshold; the threshold is compatible with the framework but not derived from first principles.

Status: Calibrated / Conditional | *Formalizability:* $\vdash_P$ | *Depends on:* Resolution | *Stage(s):* 070, 074

Chapter 11

Magnitude and Closure Theorems (the S77–S86 arc)

This arc is the quantitative core — and the most subtle. It must be read with its statuses. **Theorem 11.1**

(Hierarchy Localization). The critical invariants are of order unity ($\eta = \frac{1}{4}$, $\rho_s = \frac{2}{\pi}$) and do not generate the observed hierarchy; the problem is localized outside the critical sector.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Criticality | *Stage(s):* 077, 080

Theorem 11.2 (Dimensional Transmutation). Scale-breaking by contact closure re-opens the magnitudes; dimensional transmutation can generate emergent hierarchy ratios.

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Localization | *Stage(s):* 078, 079

Theorem 11.3 (Distance-to-Criticality Attractor). The distance to criticality t is a self-organized attractor $t^* = \sqrt{y/\gamma}$, reducing the question to the feedback strength γ .

Status: Derived | *Formalizability:* $\vdash_P$ | *Depends on:* Criticality | *Stage(s):* 081

Theorem 11.4 (Action-Derived Feedback Form). The *form* of the feedback derives from the action ($F = -\delta S/\delta\theta$); but its *magnitude* (ratio K/γ) is not fixed by that form alone.

Status: Derived (form) / Open (magnitude) | *Formalizability:* $\vdash_P$ / $\vdash_L$ | *Depends on:* Substrate action | *Stage(s):* 082–085

Equation rejoined: $S_{\text{BST}}[\theta; r] = \int \mathcal{L}_{\text{BSF}} d^d x dt$, $F_r = -\delta S/\delta\theta$.

Theorem 11.5 (Closure Coefficient Forced at the KT Fixed Point). The autonomous KT flow in which the substrate sits has a *universal* fixed point $K_R^* = 2/\pi$: the recursion “is this coefficient really free?” terminates at a *forced* invariant. But $K^* = 2/\pi$ is of order unity $\Rightarrow$ it does not generate the hierarchy.

Status: Forced (the coefficient) | *Formalizability:* $\vdash_P$ | *Depends on:* Criticality, Feedback | *Stage(s):* 086

Equation rejoined: $K_R^* = 2/\pi$ (ordered side); disordered side: runaway (vortex unbinding).

Theorem 11.6 (Physical Magnitudes). The observed values of the masses, G and Λ require calibration; existence derived, values calibrated.

Status: Calibrated | *Formalizability:* $\vdash_P$ | *Depends on:* Matter, Gravity | *Stage(s):* 077

Equation rejoined: $G_{\text{eff}} \sim \text{curvature/energy}$; $\Lambda_{\text{eff}} \sim L_{\text{IR}}^{-2}$ (S064).

Theorem 11.7 (Residual Closure). Every internal coefficient is forced and the ontology is unified; there remain exactly (i) one off-critical datum (the bare off-critical coupling) and (ii) a meta-axiomatic wall. The framework is *mapped*, not *finished*.

Status: Open (the hierarchy) | *Formalizability:* $\vdash_L$ | *Depends on:* The whole arc | *Stage(s):* 086

Chapter 12

Methodological Theorems

Not derived from the framework: these are the constraints it imposes on itself. Formalizable as logical conditions on admissible reconstructions. **Theorem 12.1 (Necessity).** A construct is admissible

only if its removal breaks the explanatory chain.

Status: Methodological | *Formalizability:* $\vdash_L$ | *Depends on:* — | *Stage(s):* —

Theorem 12.2 (Non-Cheating). Derivation goes hypotheses $\rightarrow$ consequences; *a posteriori* insertion of a desired result is forbidden.

Status: Methodological | *Formalizability:* $\vdash_L$ | *Depends on:* — | *Stage(s):* 073, 078

Theorem 12.3 (Persistence, Closure, Reproducibility). An admissible reconstruction must persist under the dynamics, maintain its closure, and remain reproducible when numerical.

Status: Methodological | *Formalizability:* $\vdash_L / \vdash_P$ | *Depends on:* Necessity, Non-Cheating | *Stage(s):* 083–085

Chapter 13

Synthesis and Formalization Program

13.1 What the compendium establishes

A coherent skeleton: a forced/derived ontological chain; a theory of resolution in which the latent is a *necessary* consequence; a physical sector that *rejoins* established physics of electromagnetism (Coulomb, Maxwell, magnetism), of fermions (Dirac, spin- $\frac{1}{2}$, Clifford), of quantum mechanics (quantum potential, de Broglie, Schrödinger), of relativity and gravity (dispersion, Newton, conical Einstein 2+1, $\gamma = 1$); Standard Model inputs explicitly *posited* (colour Z_3 , generations); and a magnitude arc whose core (S086) shows the closure coefficient to be *forced* at the KT fixed point $2/\pi$.

13.2 What the compendium does not establish

Neither the empirical truth of BST, nor a derivation of the observed magnitudes, nor crossing the off-critical datum and the meta-axiomatic wall, nor a proof of the cognitive layer (Exploratory). The convergence of the two doors attests to coherence, not experimental validation.

13.3 The Lean target

The $\vdash_L$ subset is formalizable: the ontological theorems, the resolution theorems (especially *Latent Necessity*), Reconstruction, *Residual Closure*, and the methodological criteria. The work consists in *fixing the definitions* of the primitives — resolution as a family of projections, observable as a quotient, latent as a non-empty complement, persistence as coherence across projections, closure as a fixed-point condition. The $\vdash_P$ theorems belong to the mathematics of known physics; the $\vdash_E$ ones are not formalizable as they stand.

13.4 Final admissible statement

A successful Lean verification will license this — and nothing stronger: “the logical scaffolding of BST’s core is machine-checked for internal consistency, given its definitions and axioms.” A real and bounded achievement; it will say nothing about BST’s correspondence with nature, which remains an open empirical question.